

C. Dijkstra?

time limit per test: 1 second

memory limit per test: 64 megabytes

You are given a weighted undirected graph. The vertices are enumerated from 1 to n . Your task is to find the shortest path between the vertex 1 and the vertex n .

Input

The first line contains two integers n and m ($2 \leq n \leq 10^5$, $0 \leq m \leq 10^5$), where n is the number of vertices and m is the number of edges. Following m lines contain one edge each in form a_i, b_i and w_i ($1 \leq a_i, b_i \leq n$, $1 \leq w_i \leq 10^6$), where a_i, b_i are edge endpoints and w_i is the length of the edge.

It is possible that the graph has loops and multiple edges between pair of vertices.

Output

Write the only integer -1 in case of no path. Write the shortest path in opposite case. If there are many solutions, print any of them.

Examples

input	Copy
<pre> 5 6 1 2 2 2 5 5 2 3 4 1 4 1 4 3 3 3 5 1 </pre>	
output	Copy
<pre> 1 4 3 5 </pre>	

input	Copy
<pre> 5 6 1 2 2 2 5 5 2 3 4 1 4 1 4 3 3 3 5 1 </pre>	
output	Copy
<pre> 1 4 3 5 </pre>	

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